

William C. Dawn
CURRICULUM VITAE — OCTOBER 2026

CONTACT INFORMATION

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EDUCATION

North Carolina State University, Ph.D. Nuclear Engineering 2022
Dissertation: *Multiphysics Modeling of Microreactors with Unstructured Mesh Neutron Transport and Exascale Computing Architectures*.
Advisor: Scott Palmtag.

North Carolina State University, M.S. Nuclear Engineering 2019
Thesis: *Simulation of Fast Reactors with the Finite Element Method and Multiphysics Models*.
Advisor: Scott Palmtag.

North Carolina State University, B.S. Nuclear Engineering 2017

RESEARCH EXPERIENCE

Idaho National Laboratory, Idaho Falls, ID 2026–Present
Computational Scientist. Computational Frameworks.
Leading reactor physics modeling for autonomous control of TRIGA reactor. Developer of FEM software.

Idaho National Laboratory, Idaho Falls, ID Summer 2019
Graduate Research Assistant. Reactor Physics Modeling & Simulation.
Added ISOTXS and MC²-3 interface to Griffin. Analyzed anisotropic scattering in fast reactors.

Oak Ridge National Laboratory, Oak Ridge, TN Summer 2017
Graduate Research Assistant. Reactor Multiphysics.
Used MC²-3 and ISOTXS file reader to analyze Molten Chloride Fast Reactor (MCFR) with MPACT.

Consortium for Advanced Simulation of LWRs (CASL), Raleigh, NC 2016–2019
Undergraduate & Graduate Research Scholar. Physics Integration.
Verification & Validation of MPACT cross sections. Runtime improvements of CTF steam tables.

North Carolina State University, Raleigh, NC 2017–2022
Graduate Student. Nuclear Engineering University Program (NEUP) Fellow.

PROFESSIONAL EXPERIENCE

Studsvik Scandpower, Inc., Idaho Falls, ID 2022–2026
Senior Nuclear Engineer. Methods.
Lead developer of new Peacock Monte Carlo code.
Developer for CASMO/SIMULATE with an emphasis on transient analysis.

GE Hitachi Nuclear Americas, LLC, Wilmington, NC Summer 2015 & 2016
Edison Engineering Intern. LOCA & Containment. PRISM.
LOCA analysis with TRACG. Sodium cold trap design for PRISM.

NCSU PULSTAR Reactor, Raleigh, NC 2014–2017
Reactor Operator. Nuclear Reactor Program.
Licensed by U.S. NRC to operate 1 MW pool-type research reactor.

TEACHING EXPERIENCE

- University of Idaho**, Nuclear Engineering & Industrial Management Dept. Fall 2026
Nuclear Criticality Safety — NE 5350
Instructor of record. Graduate course. Hybrid format. Developed syllabus, homework, and projects.
- Idaho National Laboratory** August 5, 2026
Nuclear 101
Developed and delivered introductory course for non-nuclear engineers.
- North Carolina State University**, Nuclear Engineering Dept. Fall 2019
Fundamentals of Nuclear Engineering — NE 301
Guest lecturer. Undergraduate course. Lectures on neutron diffusion, depletion, and xenon.
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MENTORING EXPERIENCE

- University of Michigan**, Ann Arbor, MI September 2026–Present
Working with Prof. Brendan Kocuhunas advising a student researching FEM and GSP_N.
- Idaho National Laboratory**, Idaho Falls, ID Summer 2026
Mentored graduate student intern resulting in conference paper submission.
- Studsvik Scandpower, Inc.**, Idaho Falls, ID Summer 2024 & 2025
Re-established internship program. Mentored interns resulting in four conference papers.
- North Carolina State University**, Raleigh, NC 2020–Present
Developed LUPINE multi-physics suite for SFRs. Mentor students continuing to develop LUPINE.
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REFEREED JOURNAL ARTICLES

- [1] **William C. Dawn**, Charles Wemple, Tung D. C. Nguyen, Joshua Hykes, Petri Forslund Guimarães, and Rodolfo M. Ferrer. “Peacock: A Monte Carlo Code for Reactor Physics.” In: *Nuclear Science and Engineering* (2026). Accepted.
- [2] **William C. Dawn**. “The Even Moment Method for Solving the P_N form of the Neutron Transport Equation and Challenges in Computing Multi-Group Transport Cross Sections with Anisotropic Scattering.” In: *Nuclear Science and Engineering* (2026). doi: 10.1080/00295639.2026.2706875.
- [3] **William C. Dawn**, Gerardo Grandi, and Tamer Bahadir. “Validation of SIMULATE5-K and CASMO5 with the SPERT-III E-Core.” In: *Frontiers in Nuclear Engineering* (2026). doi: 10.3389/fnuen.2026.1771859.
- [4] **William C. Dawn**. “Method of Manufactured Solutions for the Neutron Kinetics Equations.” In: *Nuclear Science and Engineering* 200.8 (2026). doi: 10.1080/00295639.2025.2551471.
- [5] **William C. Dawn**. “A Benchmark for Neutron Kinetics Methods in Hexagonal Geometry.” In: *Nuclear Science and Engineering* 199.5 (2025). doi: 10.1080/00295639.2024.2396173.
- [6] **William C. Dawn**, Gerardo Grandi, and Tamer Bahadir. “Development and Benchmarking of SIMULATE5-K-VVER.” In: *Annals of Nuclear Energy* 195 (2024). doi: 10.1016/j.anucene.2023.110146.
- [7] **William C. Dawn** and Scott Palmtag. “Solving the Neutron Transport Equation for Microreactor Modeling Using Unstructured Meshes and Exascale Computing Architectures.” In: *Nuclear Science and Engineering* 197.12 (2023). doi: 10.1080/00295639.2023.2189510.
- [8] **William C. Dawn** and Scott Palmtag. “A Multiphysics Simulation Suite for Liquid Metal-Cooled Fast Reactors.” In: *Annals of Nuclear Energy* 159 (2021). doi: 10.1016/j.anucene.2021.108213.

- [9] Robert Salko, Aaron Wysocki, Taylor Blyth, Aysenur Toptan, Jianwei Hu, Vineet Kumar, Chris Dances, **William Dawn**, et al. “CTF: A Modernized, Production-Level, Thermal Hydraulic Solver for the Solution of Industry-Relevant Challenge Problems in Pressurized Water Reactors.” In: *Nuclear Engineering and Design* 397 (2022). doi: 10.1016/j.nucengdes.2022.111927.

REFEREED CONFERENCE PROCEEDINGS

- [10] **William C. Dawn** and Charles Wemple. “Temperature Interpolation of Cross Sections in the Peacock Monte Carlo Code.” In: *Proceedings of PHYSOR 2026*. Torino, Italy, Apr. 2026. doi: 10.5281/zenodo.20803239.
- [11] William Madsen and **William C. Dawn**. “Analysis of the Neutronic Effects of the Axial Twist of Helical Cruciform Fuel Rods Using Peacock.” In: *Proceedings of PHYSOR 2026*. Torino, Italy, Apr. 2026. doi: 10.5281/zenodo.20803323.
- [12] **William C. Dawn**, William Madsen, and Charles Wemple. “Investigation of Chlorine Cross Sections in Molten Chloride Fast Reactors Using Peacock.” In: *Proceedings of PHYSOR 2026*. Torino, Italy, Apr. 2026. doi: 10.5281/zenodo.20803857.
- [13] Charles A. Wemple, William Madsen, and **William C. Dawn**. “Benchmark Testing of the Peacock Monte Carlo Code System.” In: *Proceedings of PHYSOR 2026*. Torino, Italy, Apr. 2026. doi: 10.5281/zenodo.20803880.
- [14] Tung D. C. Nguyen, **William C. Dawn**, Charles A. Wemple, Tamer Bahadir, Rodolfo M. Ferrer, Joshua Hykes, and Joel Rhodes III. “Multi-Group Cross Section Generation with Peacock for Fast Reactor Applications.” In: *Proceedings of PHYSOR 2026*. Torino, Italy, Apr. 2026. doi: 10.5281/zenodo.20803855.
- [15] **William C. Dawn** and Charles Wemple. “Peacock: A Monte Carlo Transport Code with Depletion for Reactor Modeling.” In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.
- [16] C. A. Wemple, R. M Ferrer, T. T. Simeonov, J. M. Hykes, **W. C. Dawn**, and J. D. Rhodes. “Nuclear Data Usage at Studsvik Scandpower and Future Needs from ENDF/B-IX.” In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.
- [17] Rodolfo Ferrer, **William C. Dawn**, and Tamer Bahadir. “Verification of CMS5 for Higher Enrichment and High Burnup PWR Core Design.” In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.
- [18] **William C. Dawn**. “Peacock: A Monte Carlo Code with Shared Memory Parallelism for Consumer-Grade Computing Architectures.” In: *Proceedings of M&C 2025*. Denver, CO, Apr. 2025.
- [19] **William C. Dawn**, Charles Wemple, Joshua Hykes, Rodolfo M. Ferrer, and Joel Rhodes III. “Peacock: Development of a New Continuous Energy Monte Carlo Transport Code.” In: *Transactions of the American Nuclear Society*. Vol. 131. Nov. 2024.
- [20] **William C. Dawn**, Gerardo Grandi, and Tamer Bahadir. “Initial Validation of SIMULATE5-K and the CMS5 Reactor Modeling Suite with the SPERT-III Experiments.” In: *Proceedings of PHYSOR 2024*. San Francisco, CA, Apr. 2024. doi: 10.13182/PHYSOR24-43302.
- [21] **William C. Dawn** and Tamer Bahadir. “Development and Benchmarking of Transient Nodal Code SIMULATE5-K Neutron Kinetics Solver for VVERs and Hexagonal Geometries.” In: *Proceedings of M&C 2023*. Niagara Falls, Ontario, Aug. 2023.

- [22] Timothy M. Kiefer, **William C. Dawn**, Khaldoon Al-Dawood, and Scott Palmtag. “Control Rod Modeling in Liquid Metal-Cooled Fast Reactors.” In: *Proceedings of PHYSOR 2022*. Pittsburgh, PA, May 2022. DOI: 10.13182/PHYSOR22-37649.
- [23] **William C. Dawn** and Scott Palmtag. “Simplified Thermal Expansion Modeling for Liquid Metal-Cooled Fast Reactors.” In: *Proceedings of M&C 2021*. Raleigh, NC, Oct. 2021.
- [24] Khaldoon A. Al-Dawood, **William C. Dawn**, and Scott Palmtag. “Multiphysics Simulation of Uranium-Nitride Fueled Lead-Cooled Fast Reactor.” In: *Proceedings of M&C 2021*. Raleigh, NC, Oct. 2021.
- [25] Scott Palmtag, **William C. Dawn**, and Chase Lawing. “Fast Reactor Depletion Methods in LUPINE.” In: *Proceedings of M&C 2021*. Raleigh, NC, Oct. 2021.
- [26] **William C. Dawn** and Scott Palmtag. “A Multiphysics Simulation Suite for Sodium Cooled Fast Reactors.” In: *Proceedings of PHYSOR 2020*. Cambridge, UK, Apr. 2020. DOI: 10.1051/epjconf/202124706019.

TECHNICAL REPORTS

- [27] **William C. Dawn**, Javier Ortensi, Mark D. Dehart, and Scott P. Palmtag. *Comparison of Higher-Order Neutron Scattering Cross Sections*. Tech. rep. INL/EXT-19-54899. Idaho National Laboratory, 2019. DOI: 10.2172/1593864.
- [28] **William C. Dawn**. *An Analytic Benchmark for the Solution to the Isotopic Fission Spectrum Mixture Problem*. Tech. rep. INL/EXT-19-54998. Idaho National Laboratory, 2019. DOI: 10.2172/1593873.

DISSERTATION AND THESIS

- [29] **William C. Dawn**. “Multiphysics Modeling of Microreactors with Unstructured Mesh Neutron Transport and Exascale Computing Architectures.” Ph.D. Dissertation. North Carolina State University, 2022.
- [30] **William C. Dawn**. “Simulation of Fast Reactors with the Finite Element Method and Multiphysics Models.” M.S. Thesis. North Carolina State University, 2019.

PATENTS

- [31] Eric P. Loewen, James P. Sineath, Dean D. Molinaro, **William C. Dawn**, Robin D. Sprague, Theron D. Marshall, and Joel P. Melito. “Intermixing Feedwater Sparger Nozzles and Methods for Using the Same in Nuclear Reactors.” Pat. US20180277265A1 (Wilmington, NC). Feb. 2020.
- [32] Eric P. Loewen, James P. Sineath, Dean D. Molinaro, **William C. Dawn**, William J. Garcia, Oscar L. Meek, and Patrick K. Day. “Acoustic Flowmeter and Methods of Using Same.” Pat. US20180277267A1 (Wilmington, NC). May 2020.
- [33] James P. Sineath, Dean D. Molinaro, **William C. Dawn**, and Eric P. Loewen. “Systems and Methods for Airflow Control in Reactor Passive Decay Heat Removal.” Pat. US10937557B2 (Wilmington, NC). Mar. 2021.

SELECTED INVITED TALKS

North Carolina State University, Nuclear Engineering Dept. — Raleigh, NC September 10, 2026
Seminar. “Autonomous Operations of NRAD.”
American Nuclear Society, Summer Conference — Chicago, IL June 16, 2025
Panelist. Roundtable on Current Issues in Computational Methods. Topic: “Side Projects that Took Off.”

TECHNICAL SKILLS

Languages C++, Fortran, C, Python, MATLAB, Bash, $\LaTeX$
Simulation Packages CASMO, SIMULATE, MFEM, MOOSE, OpenMC, MCNP, Serpent, MPACT, CTF, VERA, DIF3D, MC²-3, REBUS
General Proficiencies Git, GitHub, GitLab, HDF5, Make, CMake

HONORS & AWARDS

American Nuclear Society — Alan F. Henry/Paul A. Greebler Graduate Scholarship 2021
U.S. Department of Energy — NEUP Graduate Fellowship 2017–2020
North Carolina State University — College of Engineering Master’s Scholar of the Year 2019
U.S. Department of Energy — NEUP Undergraduate Scholarship 2014–2017
American Nuclear Society — Undergraduate Scholarship 2014–2018

SERVICE

American Nuclear Society, Member 2014–Present
Reactor Physics & Mathematics and Computation divisions
Early Career Researchers Association (ECRA) at INL, Member 2026–Present
Cross Section Evaluation Working Group (CSEWG), Member 2025–Present
Annals of Nuclear Energy, Referee 2022–Present
Nuclear Science and Engineering, Referee 2022–Present
Nuclear Engineering and Design, Referee 2022–Present